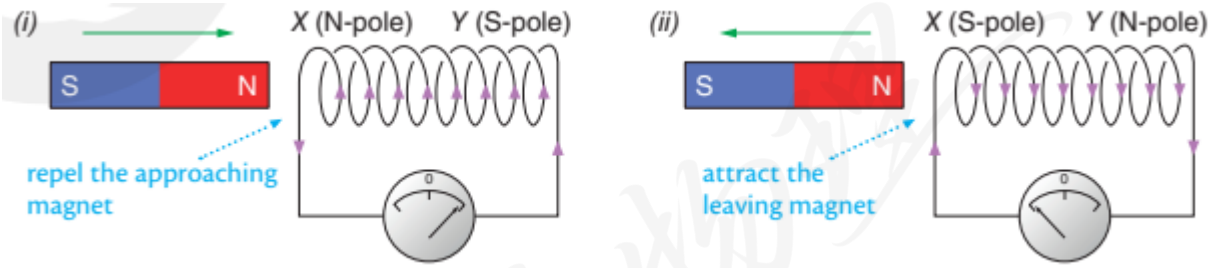
4.7.1 Conditions for electromagnetic induction

Changing magnetic flux linkage induces an e.m.f.. Faster change gives a greater magnitude. Changes in field, coil area or orientation can cause induction, so mechanical motion is not essential. An induced e.m.f. drives a sustained current only when a closed conducting path is present.

Lenz's law

The magnetic effect of induced current opposes the flux change causing it. When original flux increases, the induced field opposes the increase. When original flux decreases, the induced field points in the original direction to oppose the decrease. First identify the flux change, then find the induced field direction and use the right-hand rule to determine current direction.

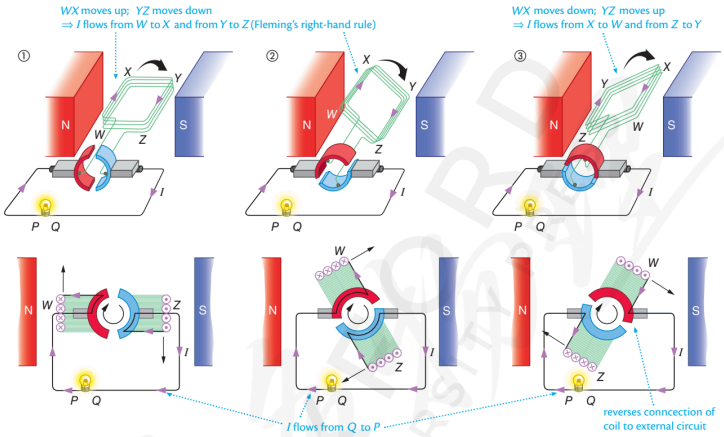


The induced field opposes the flux change caused by an approaching or receding magnet.

4.7.2 The d.c. generator

A d.c. generator converts mechanical energy into electrical energy. Each coil end connects to one half of a split-ring commutator. The two halves are mutually insulated and connect to the external circuit through carbon brushes.

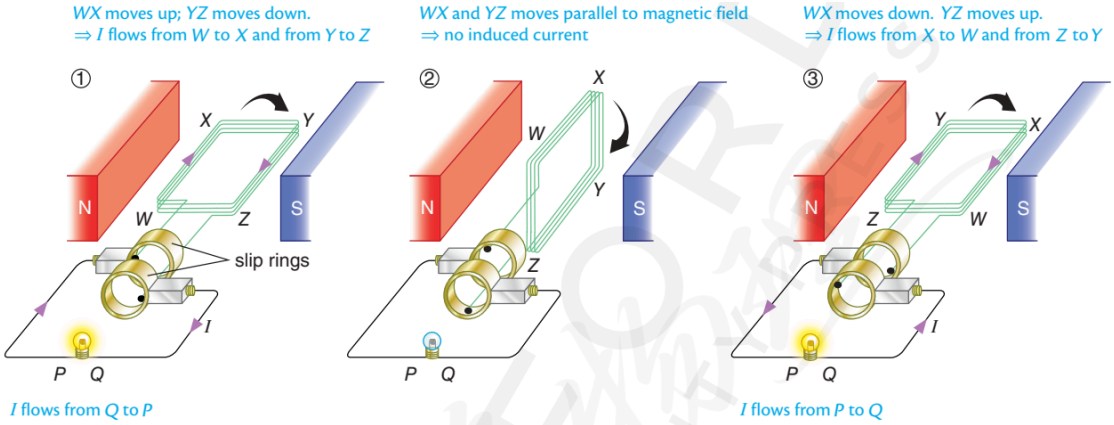
As the coil rotates in a magnetic field, its induced e.m.f. reverses every half-turn. At the same time, the commutator halves exchange brushes, swapping the coil ends' external connections. The current in the coil reverses, but the external current keeps the same direction. Its magnitude varies, so the output is pulsating d.c.



The commutator swaps connections every half-turn; coil current reverses while external current still flows from Q to P.

4.7.3 The a.c. generator

An a.c. generator converts mechanical energy into electrical energy. Each coil end connects to one slip ring. These are two complete metal rings, insulated from each other, each connected to the external circuit through its own carbon brush. As the coil rotates in a magnetic field, its induced e.m.f. reverses every half-turn. Slip rings maintain each coil end's connection to its respective brush, without swapping connections as a commutator does. The current in both the coil and external circuit therefore reverses every half-turn, producing a.c.



Slip rings maintain the connections; after half a turn, external current changes from Q → P to P → Q.

4.7.4 Eddy currents

Changing flux through conducting loops within a bulk conductor induces circulating eddy currents. Their magnetic effects oppose the change and can provide magnetic braking. Their resistive heating is used in induction heating. Laminating transformer cores with insulated thin sheets restricts large circulating paths and reduces these losses.

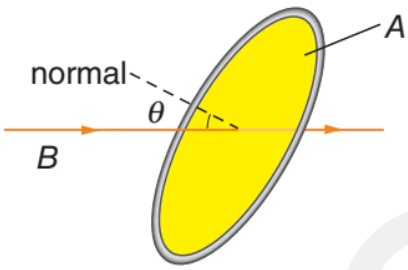


Fig 7.3c *B making an angle θ with the normal of the coil.*

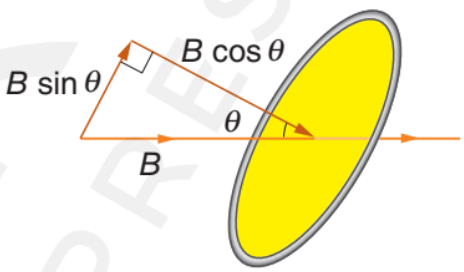


Fig 7.3d *Components of B parallel and perpendicular to the normal of the coil.*

θ is the angle between B and the area normal. Only B cos θ passes normally through the area; flux is zero when the field is parallel to the surface.

4.7.5 Magnetic flux and area orientation

Magnetic flux through a flat surface in a uniform field equals flux density times the area projected perpendicular to the field. Measure the angle from the surface normal: flux is largest when the field is perpendicular to the plane and zero when parallel. Flux, measured in Wb, differs from flux density, measured in T.

$$\Phi = BA\cos\theta$$
$$1\ \mathrm{T} = 1\ \mathrm{Wb\ m^{-2}}$$

Uniform field through a flat area: Φ (Wb); B (T); A (m²); θ: angle between B and the surface NORMAL, not the plane.

Magnetic flux density equals flux per perpendicular area for a uniform field.

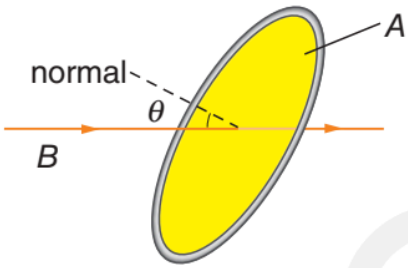


Fig 7.3c *B making an angle θ with the normal of the coil.*

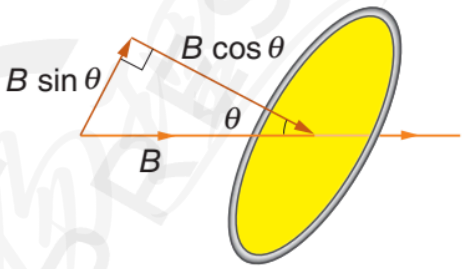


Fig 7.3d *Components of B parallel and perpendicular to the normal of the coil.*

θ is the angle between B and the area normal. Only B cos θ passes normally through the area; flux is zero when the field is parallel to the surface.

4.7.6 Magnetic flux linkage

Flux linkage is the sum of the flux linked by each turn. It equals number of turns times flux per turn only when every turn links the same flux. Increasing turns increases the induced e.m.f. for the same per-turn flux change.

$$\Psi = N\Phi$$

Ψ: magnetic flux linkage (Wb turn); N: turns; Φ: flux through each turn (Wb). All turns link the same flux.

Faraday's law

Induced e.m.f. magnitude is the rate of change of flux linkage. Average e.m.f. uses linkage change divided by elapsed time. Instantaneous e.m.f. follows the tangent slope of a flux-linkage-time graph. Take the absolute value when only magnitude is needed. For signed values, choose consistent normal and winding directions. The minus sign expresses Lenz's law.

$$\varepsilon_{\text{av}} = -\frac{\Delta\Phi}{\Delta t} \quad (N = 1)$$

$$\varepsilon_{\text{av}} = -N\frac{\Delta\Phi}{\Delta t}$$

ε: average induced e.m.f. (V); Φ: flux per turn (Wb); N: fixed turns; Δt: s. Minus sign expresses Lenz's law. Instantaneous e.m.f. uses the instantaneous rate of change.

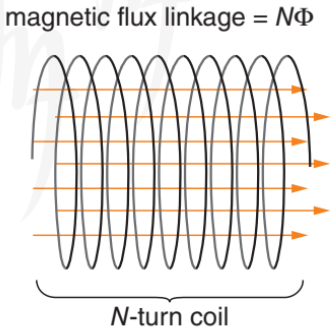
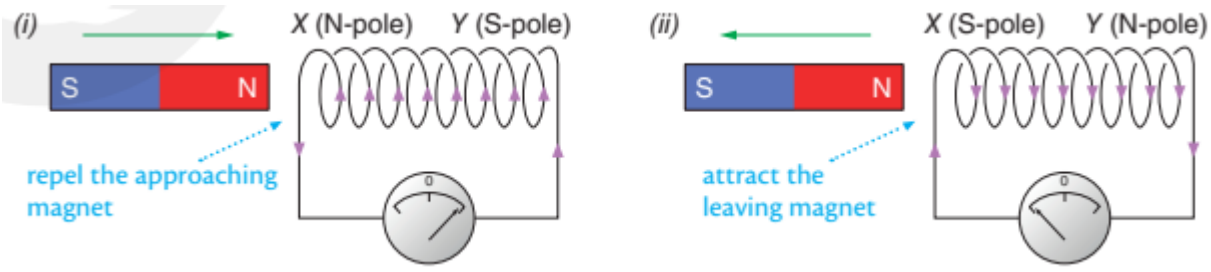


Fig 7.3e *Magnetic flux linkage through an N-turn coil.*

Each of N turns links flux Φ, so total flux linkage is NΦ. A change in this total determines the induced e.m.f.



The induced field opposes the flux change caused by an approaching or receding magnet.